MECH321
Dynamics of Engineering Systems

Lecture 6
Physical Modelling and State Space

Prof. Zengxi (Stephen) Pan

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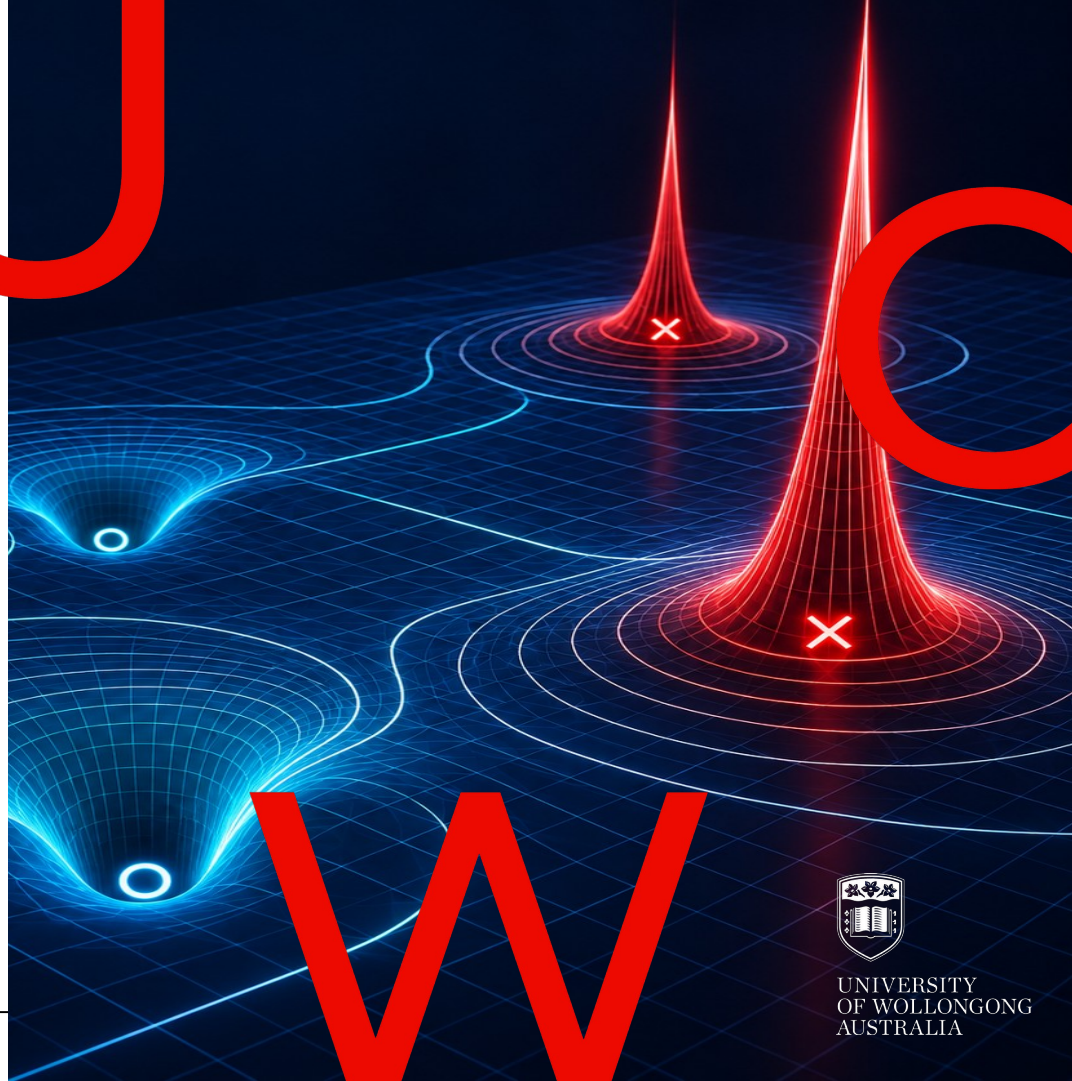


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PART 1

Circuits and mechanical analogies

01 RC and RLC circuit models

02 Impedance and cascade loading

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Modelling physical systems

Mechanical	Electrical	Electromechanical	Fluid / thermal
Lectures 1–4	RC, RLC circuits	DC motor and robot joint	Tank, valve, sensor

Physical laws → **ODE + order** → **Transfer function**

Newton, KVL, balances

Count independent stores

Zero-state input–output

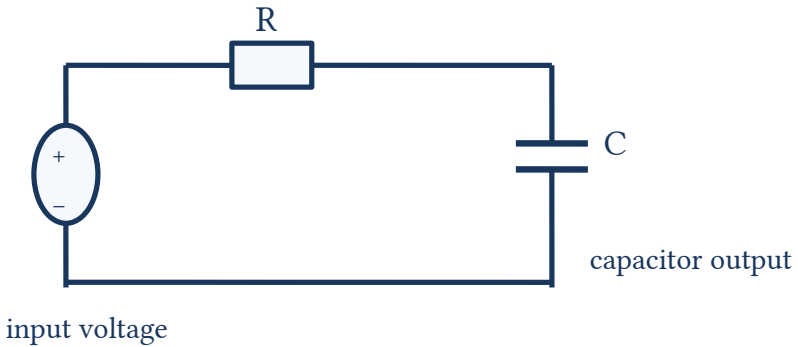
↘ **State space**

Internal states + inputs

RC circuit: current balance to transfer function

EXAMPLE

Capacitor voltage is the single independent state



KVL → capacitor law → transfer function

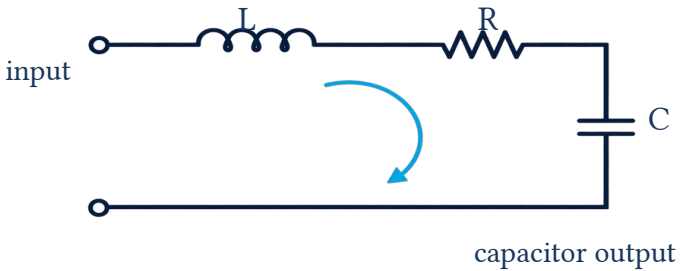
$$Ri + \frac{1}{C}\int idt = e_i$$

$$RI(s) + \frac{1}{Cs}I(s) = E_i(s)$$

$$\frac{E_o(s)}{E_i(s)} = \frac{(1/Cs)I(s)}{RI(s) + (1/Cs)I(s)} = \frac{1}{RCs+1}$$

If the capacitor initially holds charge, retain the initial-condition term; $G(s)$ alone does not describe that free response.

RLC Transfer Function



MODEL → ODE → LAPLACE → OUTPUT

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int idt = e_i$$

$$LsI(s) + RI(s) + \frac{1}{Cs}I(s) = E_i(s)$$

$$\frac{E_o(s)}{E_i(s)} = \frac{(1/Cs)I(s)}{LsI(s) + RI(s) + (1/Cs)I(s)} = \frac{1}{LCs^2 + RCs + 1}$$

1 Series current through R, L and C.
KVL; capacitor-voltage output.
3 Zero-state transform: integral $1/s$; derivative s.
4 Multiply by Cs. Independent L and C stores: s^2 denominator.

Denominator coefficients: LC, RC, 1.

Worked circuit: series RLC

EXAMPLE

$R = 4\ \Omega, L = 1\ \text{H}, C = 0.25\ \text{F}$

Input $v(t)$, output capacitor voltage $v_C(t)$.

KVL around the loop:

$$\frac{L di}{dt} + Ri + v_C = v$$

The capacitor links current to the output:

$$i = \frac{C dv_C}{dt}$$

Eliminate i and collect

$$LC\ddot{v}_C + RC\dot{v}_C + v_C = v$$

$$0.25\ddot{v}_C + \dot{v}_C + v_C = v$$

$$\frac{V_C(s)}{V(s)} = \frac{1}{0.25s^2 + s + 1}$$

$\omega_n = 2$ rad/s, $\zeta = 1 \rightarrow$ **critically damped**

Read ζ and ω_n using the standard form.

Complex Impedance

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Use element impedances for transfer functions and sinusoidal gain and phase

Zero initial conditions: each element obeys $V(s) = Z(s)I(s)$.

Element	Time-domain law	Impedance $Z(s)$
Resistor	$v = Ri$	$Z_R = R$
Inductor	$v = L\dfrac{di}{dt}$	$Z_L = sL$
Capacitor	$i = C\dfrac{dv}{dt}$	$Z_C = \dfrac{1}{sC}$

For a series loop, add impedances. Select the output using a voltage divider.

Nonzero initial energy requires additional source terms.

Worked impedance solution: series RLC

EXAMPLE

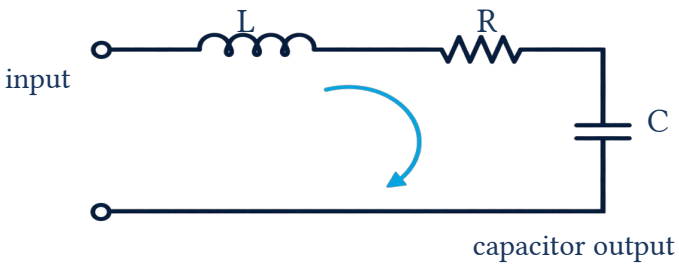
Question Series R–L–C; source $u(t)$, output capacitor voltage $v_C(t)$.

The same circuit and values as the preceding KVL example: $R = 4$ Ω, $L = 1$ H, $C = 0.25$ F.

$$Z_R = R, \qquad Z_L = sL, \qquad Z_C = \frac{1}{sC}$$

$$G(s) = \frac{Z_C}{Z_L + Z_R + Z_C} = \frac{1}{LCs^2 + RCs + 1}$$

$$G(s) = \frac{1}{0.25s^2 + s + 1} = \frac{4}{(s+2)^2}$$



Check: two stores give two poles. The repeated pole at −2 matches the KVL derivation.

Mechanical–electrical analogy

Role	Mechanical	Electrical	Physical meaning
Effort / flow	force / velocity	voltage / current	power-conjugate pair
Inertia	mass	inductor	energy storage
Dissipation	viscous damper	resistor	energy loss
Elasticity	spring	capacitor	stiffness maps to reciprocal capacitance

Force–voltage analogy

Matching balance equations; different physical units. Use zero initial displacement and charge.

PART 2

DC motor and first-order models

01 DC motor coupling

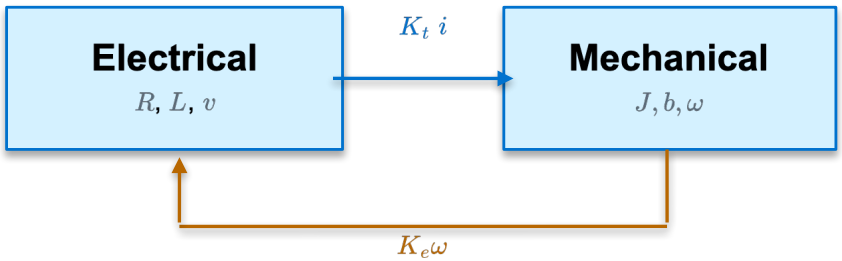
02 Motor reduction

03 Thermal extension

The DC motor

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Two subsystems, coupled in both directions



The three equations

$$\frac{Ldi}{dt} + Ri + K_e\omega = v$$

$$T = K_t i$$

$$\frac{Jd\omega}{dt} + b\omega = T$$

Eliminate i and T:

$$\frac{\Omega(s)}{V(s)} = \frac{K_t}{(Ls+R)(Js+b) + K_tK_e}$$

Back EMF $K_e\omega$ couples speed to the electrical balance; K_tK_e appears in the denominator.

DC motor: voltage and opposing load

Eliminate current once, then separate the two speed-response paths

1 Positive load opposes positive rotation (slide 12).

$$J\frac{d\omega}{dt} + b\omega = K_t i - \tau_l(t)$$

2 Zero initial current and speed: transform both balances.

$$(Ls + R)I(s) + K_e\Omega(s) = V(s)$$

$$(Js + b)\Omega(s) = K_t I(s) - \mathcal{T}_l(s)$$

3 Solve KVL for current. Substitute into torque balance, then multiply by the electrical factor.

$$I(s) = \frac{V(s) - K_e\Omega(s)}{Ls + R}$$

$$D(s) = (Ls + R)(Js + b) + K_t K_e$$

$$D(s)\Omega(s) = K_t V(s) - (Ls + R)\mathcal{T}_l(s)$$

4 Zero the other input. Same poles; different numerators and signs.

$$G_v(s) = \frac{K_t}{D(s)} \qquad G_\tau(s) = -\frac{Ls + R}{D(s)}$$

Reduced DC motor model

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Notation: K=Kt, Kb=Ke, La=L, Ra=R and Ea=V from the preceding motor model.

1 Current and speed states. Electrical scale $\tau_e = L_a/R_a$; loaded mechanical scale $\frac{J}{b+KK_b/R_a}$.
2 Quasi-steady current requires τ_e much shorter than motion and input time scales.
3 Set $L_a \approx 0$; substitute algebraic current into torque balance: first-order speed.
Position integrates speed: one additional pole at zero.

FULL MODEL

$$\frac{\Theta(s)}{E_a(s)} = \frac{K}{s\left[L_aJs^2 + (L_ab + R_aJ)s + R_ab + KK_b\right]}$$

REDUCED POSITION TRANSFER FUNCTION

$$\frac{\Theta(s)}{E_a(s)} = \frac{K}{s\left(R_aJs + R_ab + KK_b\right)}$$

STANDARD REDUCED POSITION FORM

$$\frac{\Theta(s)}{E_a(s)} = \frac{K_m}{s\left(T_ms + 1\right)}$$

Boundary: use the full model for rapid voltage changes, current-control transients or startup.

The universal first-order model

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In the tank row, C is cross-sectional area in square metres

$$\tau\dot{y} + y = Ku \quad G(s) = \frac{K}{\tau s + 1}$$

τ — time constant, seconds K — DC gain, output units per input unit

System	Equation	Time constant
RC circuit	$RC\dot{v}_C + v_C = v_{\text{in}}$	$\tau = RC$
Tank level	$RC\dot{h} + h = Rq_{\text{in}}$	$\tau = RC$
Thermometer	$RC\dot{\theta} + \theta = \theta_a$	$\tau = RC$
Motor, L neglected	$\frac{J}{b_{\text{eff}}}\dot{\omega} + \omega = Kv$	$\tau = \frac{J}{b_{\text{eff}}}$

Motor row (armature inductance neglected):

$$b_{\text{eff}} = b + \frac{K_t K_e}{R}, \quad K = \frac{K_t}{Rb_{\text{eff}}}$$

Thermometer Model

EXAMPLE

Heat balance: storage rate = incoming heat.
2 Use $q = \frac{\theta_b - \theta}{R}$ and $C\frac{d\theta}{dt} = q$.
3 Heat balance: $RC\frac{d\theta}{dt} + \theta = \theta_b$.
4 Under zero initial deviation, transform and take output/input: $\tau = RC$, gain=1.

$$C\frac{d\theta}{dt} = q = \frac{\theta_b - \theta}{R} \quad \text{or} \quad RC\frac{d\theta}{dt} + \theta = \theta_b$$

$$RCs\Theta(s) + \Theta(s) = \Theta_b(s)$$

$$\frac{\Theta(s)}{\Theta_b(s)} = \frac{1}{RCs + 1}$$

For an initially hot thermometer, include $\theta(0)$ in the Laplace derivative. The transfer function is the zero-state input-output relation.

Worked: the thermal sensor

EXAMPLE

$12\dot{\theta} + \theta = \theta_a$

$\tau = 12$ s. Ambient steps from 20 °C to 80 °C; the sensor reads 20 °C at $t = 0$.

Find the temperature response, the reading at 12 s and the approximate 2% settling time.

$\Delta\theta = 60$° C

Answer, then check

$\theta(t) = 80 - 60e^{-\frac{t}{12}}$° C

At $t = \tau = 12$ s: 63.2 % of 60 °C is 37.9 °C

$\theta(12) = 20 + 37.9 = 57.9$° C✓

2 % settling $\approx 4\tau = 48$ s

PART 3

State space and system modes

01 State equation and solution

02 Eigenvalues and modal directions

03 Transfer-function relationship

State model and transfer-function view

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GIVEN STATE MODEL

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} x, \quad x(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

State 1 is the output. State 2 is its rate.
The input is a unit step and the initial state is zero.

TRANSFER-FUNCTION VIEW

$$G(s) = \frac{Y(s)}{U(s)}$$

One selected input–output path
Poles and zeros
Zero initial conditions

STATE-SPACE VIEW

Internal state trajectory
Initial conditions retained
Several inputs and outputs

Eigenvalues, eigenvectors and natural modes

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The poles give decay rates; eigenvectors show how the states move together

$$\det(\lambda I - A) = \lambda^2 + 3\lambda + 2$$

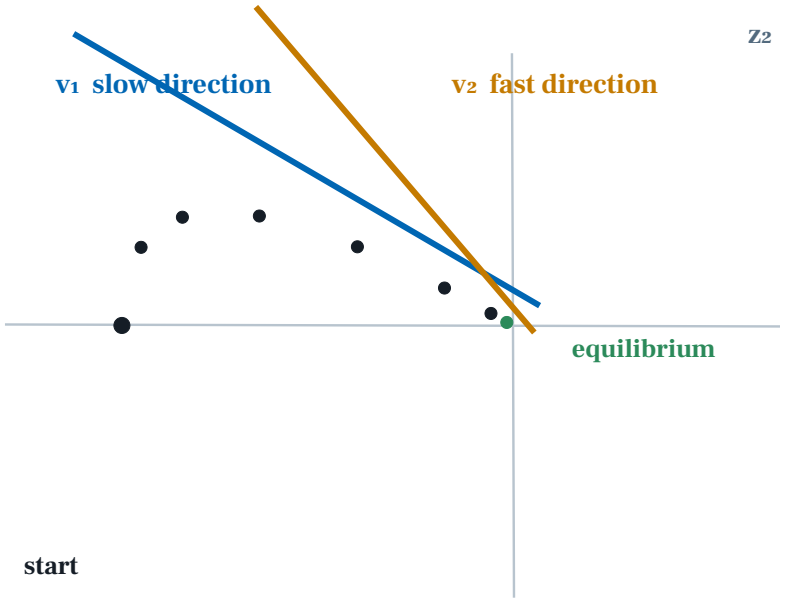
$$(\lambda + 1)(\lambda + 2) = 0$$

$$\lambda_1 = -1, \quad \lambda_2 = -2$$

$$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \qquad v_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$x_h(t) = c_1 v_1 e^{-t} + c_2 v_2 e^{-2t}$$

Each eigenvalue sets a decay rate.
Each eigenvector sets a fixed state direction.
The slower mode controls the late response.



First-state deviation from its steady value

The state points approach equilibrium as the fast mode fades and the slow mode remains.

State response and its two causes

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Initial energy and external input appear as separate terms

GENERAL SOLUTION

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau$$

INITIAL-STATE RESPONSE

$$e^{At}x(0)$$

Present with zero input.
Carries stored-state information.

INPUT-DRIVEN RESPONSE

Present from the zero initial state.
This is the transfer-function path.

$$u = \text{constant}: \quad x(t) = x_{ss} + e^{At}\left[x(0) - x_{ss}\right]$$

Unit-step response

EXAMPLE

Shift to the equilibrium, determine the modal coefficients and recover both states

SETUP AND MODAL COEFFICIENTS

$$x_{ss} = -A^{-1}B = \begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix}$$

$$z(t) = x(t) - x_{ss}, \quad z(0) = \begin{bmatrix} -\frac{1}{2} \\ 0 \end{bmatrix}$$

$$z(t) = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{-2t}$$

$$c_1 = -1, \quad c_2 = \frac{1}{2}$$

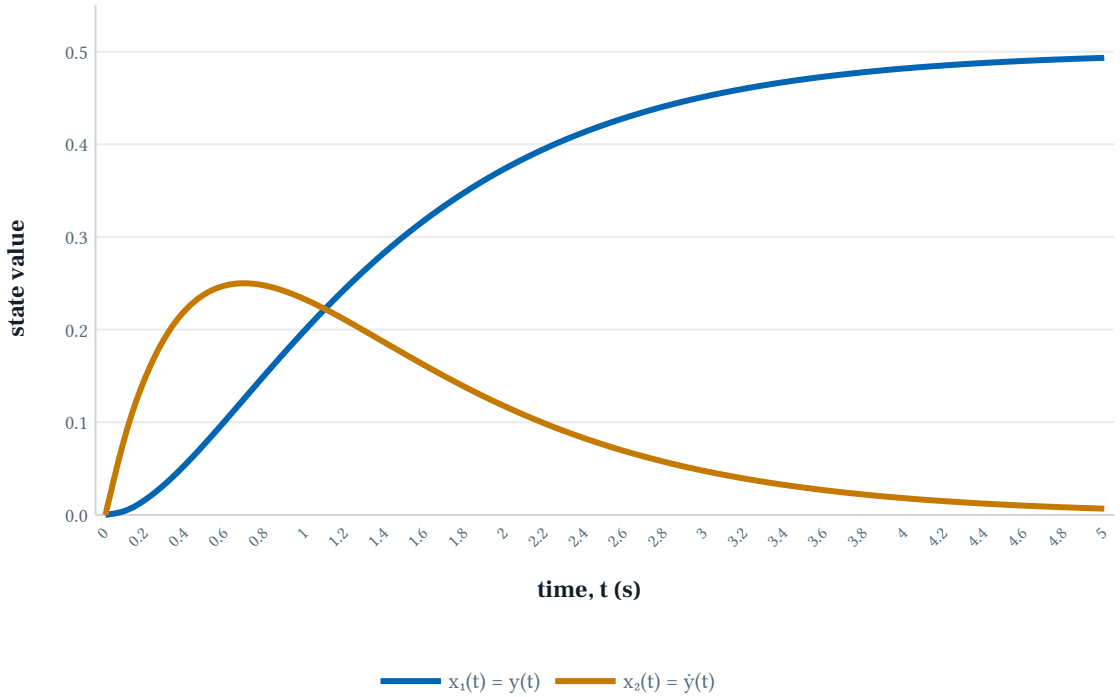
STATE TRAJECTORY

$$x_1(t) = \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}$$

$$x_2(t) = e^{-t} - e^{-2t}$$

Checks: the response starts from the zero initial state; the final state is the equilibrium calculated above.

State trajectories



READ THE CURVES

The first state rises monotonically to 0.5.

The second state peaks at 0.693 seconds, then returns to zero.

Transfer function from the matrices

The characteristic polynomial appears again in the denominator

GENERAL RELATION

$$X(s) = (sI - A)^{-1} B U(s)$$

$$G(s) = C(sI - A)^{-1} B + D$$

Set the initial state to zero.
The matrices then describe the chosen input–output channel.

TWO-STATE EXAMPLE

$$sI - A = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$\det(sI - A) = s^2 + 3s + 2$$

$$G(s) = \frac{1}{s^2 + 3s + 2}$$

For a minimal realization, transfer-function poles equal the state-matrix eigenvalues, and dimension equals order. This example has no cancellation.

State coordinates and multiple signals

STATE COORDINATES

$$x = \begin{bmatrix} y \\ \dot{y} \end{bmatrix}$$

Canonical states

$$x_p = \begin{bmatrix} i \\ \omega \end{bmatrix}$$

Physical motor states

$$x_p = Tx \quad \Longrightarrow \quad A_p = TAT^{-1}$$

A nonsingular coordinate change alters the matrices but not the external transfer function.

For a minimal model, state dimension equals transfer-function order.

MULTIPLE INPUTS AND OUTPUTS

$$u = \begin{bmatrix} v \\ \tau_L \end{bmatrix} \qquad y = \begin{bmatrix} i \\ \omega \end{bmatrix}$$

$$B \in \mathbb{R}^{n \times m}, \quad C \in \mathbb{R}^{p \times n}$$

B connects inputs to state derivatives; C selects measured outputs.

$$G(s) = [G_{ij}(s)]$$

Multiple channels form a transfer-function matrix.

Transfer function and state space

CORE

TRANSFER FUNCTION

Selected input–output path, with zero initial state
Poles, zeros and frequency response for loop shaping

STATE SPACE

Internal trajectories, with initial conditions explicit

Several inputs and outputs in one model
Supports state estimation and feedback

OPTIONAL EXTENSION: CONTROLLABILITY AND OBSERVABILITY

Can the input move every state direction?

$$\mathbf{C} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}$$

$$\operatorname{rank}(\mathbf{C}) = n$$

Can the outputs reveal every state direction?

$$\mathbf{O} = \begin{bmatrix} C^T & (CA)^T & \cdots & \left(CA^{n-1}\right)^T \end{bmatrix}^T$$

$$\operatorname{rank}(\mathbf{O}) = n$$


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